DSE Bullet Series – Information Processing and Data Control (Solutions 10)

Name: _____ ( ) Class:____ Date: _____

1. Past Questions (2012–2025) #Wrong input means wrong output

Year	Past Question	Focus / Marks
2012	Multiple-choice question on check digits	Detect some errors only; cannot correct them
2016	Features of the information age	Wide circulation of information; online information may be inaccurate
2021	Extract customer data for invitations; enter password twice	Filter/mail merge; double entry is verification
2022	Importance of information literacy in a knowledge society	Identify false information; evaluate sources and purpose
2023	Features of direct access	Locate a record directly; faster searching
2025	Information system output; data control; password verification	Input affects output; data control reduces errors

2. Key Notes #Data is not information

Information systems and processes: an information system consists of functions, data, processes, technology and people. Data is transformed into information through collection ➔ organisation ➔ analysis ➔ storage ➔ processing ➔ transmission ➔ presentation. Changing the input or the processing method can change the output.

Data and information: data are raw, unprocessed facts; information is the result of processing data so that it supports decisions. Online information may be inaccurate, so information literacy requires checking accuracy and evaluating sources.

Data organisation: a field describes one attribute; a record contains all the fields for one entity; a file is a collection of records of the same type; a database is a collection of related files shared by different applications.

File access: sequential access reads records from the beginning one by one and suits magnetic tape and batch processing; direct access locates a target record directly, is faster for searching, and is common on hard disks and flash storage.

Data control: verification ensures that input matches the source, using double entry, independent entry by two people or proofreading; validation rejects invalid input according to rules, including presence, field length, range, fixed value, format, type, consistency, uniqueness and check digit checks.

Error detection: a parity check adds a parity bit and can detect a single-bit error during transmission; a check digit is calculated from the other characters and can detect some input errors. Neither method can always correct an error.

Use C-F-B: Context (C) + data-control method (F) + benefit in reducing errors (B). Example: entering an email address twice is verification, helping detect a copying error early.
*Common mistakes: reversing data and information; confusing verification with validation; thinking a check digit can correct an error; saying only ‘direct access is faster’ without mentioning direct location of a record.

3. Similar Questions #Same process, different system

1. A library uses a computer system to record loans. After a reader taps a card, the system displays the book title, loan period and overdue fine.

(a) State one input, one process and one output of the system.

【Answer】Input: reader card/book barcode; process: check the loan record and calculate the overdue fine; output: book title, loan period and overdue fine.

(b) Explain why the library needs data control.

【Answer】It prevents incorrect or invalid loan data from producing wrong loan records, incorrect fines or system errors.

2. A school stores student data in a database with the fields Class, Class Number, Name and Telephone.

(a) State the data-organisation terms for 'Class Number', 'one student's data' and 'all students' data'.

【Answer】Class Number is a field; one student's data is a record; all students' data form a file.

(b) Compare sequential access with direct access, and give one storage device suitable for direct access.

【Answer】Direct access locates the target record directly and is faster for searching; sequential access must read from the beginning one by one and suits magnetic tape and batch processing. A hard disk or flash storage is suitable for direct access.

3. Hardy enters an identity card number, email address and age into an online form. The form asks for the email address twice and checks the identity card number's check digit.

(a) Identify the type of data control used for entering the email address twice and for the check digit, and explain one purpose of each.

【Answer】Entering the email twice is verification, ensuring the input matches the source and reducing copying errors; a check digit is a validation check that detects some input errors according to a rule.

(b) Suggest **two** validation checks that can be used for the age field.

【Answer】Range check: accept only 0 to 120; type check: accept only an integer; presence check: the field cannot be left blank. (Any two)

DSE Bullet Series – Number Systems and Two's Complement (Solutions 11)

1. Past Questions (2012–2025) #One wrong bit means a wrong answer

Year	Past Question	Focus / Marks
2012	One's complement and two's complement	Invert the bits first; add 1 for two's complement
2016	Binary/hexadecimal comparison	Use the same base; check the sign bit
2020	Two's complement to decimal; minimum bits	Subtract 256 for an 8-bit negative; $2^n \geq$ number of values
2021	Overflow in 8-bit two's complement addition	Add two numbers of the same sign; result has the opposite sign
2022	Overflow in 4-bit two's complement addition	Range −8 to 7; outside means overflow
2024	8-bit two's complement range; overflow	−128 to 127; result outside the range
2025	Two's complement addition gives 0000 0000	Values are two's complements; sum is zero

2. Key Notes #Binary, hexadecimal and two's complement

Bits and combinations: one bit has two possible values, 0 or 1; n bits give 2^n different combinations. The range of an unsigned integer is 0 to $2^n - 1$.

Base conversion: convert decimal to binary by repeated division by 2 and recording the remainders; convert binary to hexadecimal by grouping four bits from the right, e.g. 1010 0001 = A1.

Two's complement: a positive number uses ordinary binary; for a negative number, write the absolute value in binary, invert all bits and add 1. The most significant bit is the sign bit.

Range of n-bit two's complement: -2^{n-1} to $2^{n-1} - 1$; for 8 bits, −128 to 127. A result outside this range causes overflow.

Binary addition and subtraction: A − B can be performed as A + (−B) using two's complement; add bit by bit and carry 1 when the sum reaches 2.

Detecting overflow: adding two numbers of the same sign and getting a result with the opposite sign indicates overflow; a carry out of the most significant bit alone does not necessarily mean overflow.

Use C-F-B: Context (C) + number of bits/range (F) + whether the value can be represented (B). Example: 8-bit two's complement (C) can represent only −128 to 127 (F), so 128 causes overflow (B).
*Common mistakes: treating a carry from the most significant bit as overflow; forgetting to invert and then add 1 for a negative number; writing the 8-bit range as −127 to 128; grouping binary digits from the left when converting to hexadecimal.

3. Similar Questions #Same bit width, different value

1. Ka-fai stores the integers 45 and −45 in 8-bit memory.

(a) Write 45 in 8-bit binary and −45 in 8-bit **two**'s complement.

【Answer】45 = 0010 1101; invert 0010 1101 and add 1 to obtain 1101 0011 for −45.

(b) Convert the 8-bit **two**'s complement number 1110 1100 to decimal.

【Answer】The most significant bit is 1, so the value is negative; 236 − 256 = −20.

2. A computer stores integers in 8-bit **two**'s complement.

(a) Calculate 0111 0101 + 0000 1011 and state whether overflow occurs.

【Answer】The sum is 1000 0000; adding two positive numbers gives a negative result, so overflow occurs.

(b) Calculate 1000 0000 + 1111 1111 and state whether overflow occurs.

【Answer】The sum is 0111 1111; adding two negative numbers gives a positive result, so overflow occurs.

3. A school assigns a unique code to each of 40 students and stores a 16 × 16 black-and-white image in bits.

(a) What is the minimum number of bits needed for 40 unique codes? Explain.

【Answer】Six bits are needed: $2^5 = 32$ is less than 40, while $2^6 = 64$ is enough to represent 40 different codes.

(b) The image has 256 pixels. If each pixel uses 1 bit, how many bits are needed for the whole image?

【Answer】256 × 1 = 256 bits.

DSE Bullet Series – Character Encoding and Multimedia Data (Solutions 12)

1. Past Questions (2012–2024) #Text, images and sound become bits

Year	Past Question	Focus / Marks
2012	Analogue/digital conversion; pixel colours	Playback or recording involves conversion; pixels use RGB
2020	Choosing a character set; JPG/BMP	Unicode is cross-language; JPG reduces file size
2021	ASCII hexadecimal; WAV/MP3	Character encoding; WAV has better quality but larger files
2022	8×8 black-and-white image; BMP/JPG	One bit per pixel; BMP is uncompressed
2023	Unicode and Big5	Unicode covers many languages; Big5 mainly covers traditional Chinese
2024	Unicode/ASCII/GB; multimedia formats	Unicode is cross-language; format must suit the purpose

2. Key Notes #Character sets, pixels, sampling and compression

Analogue and digital: analogue data change continuously, such as temperature, sound and light intensity; digital data are represented by 0 and 1. A sensor collects an analogue signal, an analogue-to-digital converter converts it to digital data, and a digital-to-analogue converter restores it for playback. Digital data are easier to store, process, transmit and compress.

Character sets: ASCII mainly represents English; Big5 represents traditional Chinese; GB represents simplified Chinese; Unicode covers many languages, has a larger character set and can include new characters.

Images: a bitmap consists of pixels; resolution is the number of pixels, and colour depth is the number of bits per pixel. Uncompressed file size ≈ width × height × colour depth ÷ 8 bytes. Higher resolution or colour depth gives a more detailed image but a larger file; a vector image uses mathematical objects and remains sharp when enlarged.

Audio and video: audio samples the amplitude of a sound wave at a sampling rate and stores it at a bit depth; higher sampling rate and bit depth give better quality but larger files. Video consists of a sequence of frames; a higher frame rate gives smoother motion.

Formats and compression: BMP is uncompressed; JPG uses lossy compression and suits photographs; PNG uses lossless compression and supports transparency; GIF supports animation and a limited number of colours; WAV is uncompressed audio; MP3 uses lossy audio compression; AVI and MP4 are video formats. Compression reduces storage space and transmission time.

Use C-F-B: Context (C) + format/sampling feature (F) + effect (B). Example: a web page needs a transparent logo (C); PNG supports transparency (F), so no white box appears behind it (B).
*Common mistakes: confusing resolution and colour depth; assuming JPG is lossless; forgetting that file size is calculated in bytes; writing only ‘smaller file’ without naming the compression type or its effect on quality.

3. Similar Questions #Same formula, different medium

1. A bitmap image is 800 × 600 pixels and each pixel uses 24 bits.

(a) Calculate the uncompressed file size in bytes.

【Answer】 800 × 600 × 24 = 11,520,000 bits; ÷ 8 = 1,440,000 bytes.

(b) Compare bitmap and vector images when enlarged.

【Answer】 A bitmap becomes pixelated or blurred when enlarged; a vector image is defined by mathematical objects and remains sharp.

2. A 60-second stereo sound clip has a sampling rate of 44.1 kHz and a bit depth of 16 bits.

(a) Calculate the uncompressed file size in bytes.

【Answer】 44,100 × 16 × 2 × 60 = 84,672,000 bits; ÷ 8 = 10,584,000 bytes.

(b) Explain why an MP3 file is usually smaller than a WAV file, and state one trade-off.

【Answer】 MP3 uses lossy compression and removes sound data that the ear is unlikely to notice, so the file is smaller; the trade-off is that the sound quality may be lower than WAV.

3. A document contains English, traditional Chinese and simplified Chinese text.

(a) Suggest a character set that can represent all the characters, and explain why.

【Answer】 Unicode: it covers characters from many languages and can represent English, traditional Chinese and simplified Chinese together.

(b) State one difference between Big5 and Unicode.

【Answer】 Big5 mainly represents traditional Chinese and has a smaller character set; Unicode covers many languages, has a larger character set and can add new characters.

DSE Bullet Series – Spreadsheet Functions (Solutions 13)

1. Data Table #The raw material for a formula

The data are in A1:E7; row 1 contains the headings and rows 2–7 contain six students.

Name	Chinese	English	Mathematics	Class
Mei	82	91	78	Class A
Yat	65	58	72	Class B
Ka-fai	74	88	60	Class A
Ka-ching	90	76	55	Class B
Hiu-yan	58	69	83	Class A
Siu-tung	77	94	68	Class B

2. Reading a Formula #Decide the purpose, then build the formula

A formula starts with =. Text conditions must be placed in quotation marks, for example "Class A".

B2:B7 is a relative range; $B$2:$B$7 is an absolute range. Add $ to a range that must stay fixed when the formula is copied.

Functions can be nested and are evaluated from the inside out: find the maximum score first, then use that score to look up the student's name.

*Common mistakes: omitting =; leaving text conditions without quotation marks; using the wrong comparison direction; omitting $ so that a copied range shifts to the wrong cells.

3. Function Examples (1/2) #Learn the pattern from examples

Function	Purpose	Example	Result
SUM	Add a range	=SUM(B2:B7)	446
AVERAGE	Find the mean	=AVERAGE(C2:C7)	79.33
COUNT	Count numeric values	=COUNT(D2:D7)	6
COUNTIF	Count by condition	=COUNTIF(E2:E7,"Class A")	3
MAX	Find the maximum	=MAX(D2:D7)	83
MIN	Find the minimum	=MIN(B2:B7)	58
SUMIF	Add by condition	=SUMIF(E2:E7,"Class B",C2:C7)	228
RANK	Find the rank	=RANK(B2,$B$2:$B$7,0)	2
IF	Choose one of two outcomes	=IF(D5>=60,"Pass","Fail")	Fail
AND	Are all conditions true?	=AND(B2>=60,C2>=60)	TRUE
OR	Is any condition true?	=OR(D5>=60,C5>=60)	TRUE

3. Function Examples (2/2) #Learn the pattern from examples

Function	Purpose	Example	Result
NOT	Reverse TRUE/FALSE	=NOT(D5>=60)	TRUE
INT	Truncate the decimal part	=INT(87.9)	87
ROUND	Round to a number of digits	=ROUND(3.14159,2)	3.14
SQRT	Find the square root	=SQRT(144)	12
RAND	Generate a random number	=RAND()	$0 \leq x < 1$
LEFT	Take characters from the left	=LEFT("2025A123",4)	2025
RIGHT	Take characters from the right	=RIGHT("2025A123",3)	123
MID	Take characters from the middle	=MID("2025A123",5,2)	A1
LEN	Count characters	=LEN("DSE ICT")	7
FIND	Find the position of text	=FIND("ICT","DSE ICT")	5
XLOOKUP	Look up a value	=XLOOKUP("Mei",A2:A7,C2:C7)	91

4. Three Steps for a Combined Formula #DSE will not tell you which function to use

Step 1: Identify the output — should the answer be a number, text, rank or TRUE/FALSE?

Step 2: Work from the inside out — find the maximum, average or condition first, then pass it to the outer function.

Step 3: Check the ranges — do the condition, sum and return ranges refer to the correct columns? Add $ when a range must stay fixed.

Example: find the English score of the student with the highest Chinese score. Find the highest score first, then use it to look up the English score:

=XLOOKUP(MAX(B2:B7),B2:B7,C2:C7) ➔ 76

*Remember: the same result can be obtained by more than one formula. Decide on the purpose in the examination instead of memorising question types.

5. Two-Layer Formula Examples #DSE often examines a two-layer combination

Situation	Example	Result
Round the mean Chinese score to the nearest whole number	=ROUND(AVERAGE(B2:B7),0)	74
English score of the student with the highest Chinese score	=XLOOKUP(MAX(B2:B7),B2:B7,C2:C7)	76
Show "Yes" if both Chinese and English are above 75	=IF(AND(B2>75,C2>75),"Yes","No")	Yes
Number of characters before "ICT"	=FIND("ICT","DSE ICT")-1	4

6. DSE-Style Practice #Choose the function and build the formula yourself

Using the data table, write a complete formula for each situation and state its result.

Only one possible formula is shown; other formulas that produce the same result are acceptable.

1. Calculate the mean Chinese score of the whole class and round it to the nearest whole number.

   【Answer】=ROUND(AVERAGE(B2:B7),0) ➔ 74

2. Count the number of students in Class A.

   【Answer】=COUNTIF(E2:E7,"Class A") ➔ 3

3. Calculate the total English score of the students in Class B.

   【Answer】=SUMIF(E2:E7,"Class B",C2:C7) ➔ 228

4. Find the highest Mathematics score and display the Chinese score of that student.

   【Answer】=XLOOKUP(MAX(D2:D7),D2:D7,B2:B7) ➔ 58

5. Show Mei's rank in Chinese in the class (1 is the highest).

   【Answer】=RANK(B2,$B$2:$B$7,0) ➔ 2

6. Show "Yes" if Mei scores above 75 in both Chinese and English; otherwise show "No".

   【Answer】=IF(AND(B2>75,C2>75),"Yes","No") ➔ Yes

7. Count the number of students who fail Mathematics (below 60).

   【Answer】=COUNTIF(D2:D7,"<60") ➔ 1

8. Show "Pass" if Yat scores 60 or above in Mathematics; otherwise show "Fail".

   【Answer】=IF(D3>=60,"Pass","Fail") ➔ Pass

9. Find the lowest Chinese score and return the name of that student.

   【Answer】=XLOOKUP(MIN(B2:B7),B2:B7,A2:A7) ➔ Hiu-yan

10. Extract **two** characters starting from the fifth character of "2025A123".

    【Answer】=MID("2025A123",5,2) ➔ A1

11. Find where "ICT" starts in "DSE ICT" and calculate the number of characters before it.

    【Answer】=FIND("ICT","DSE ICT")-1 ➔ 4

12. Calculate the mean of Mei's Chinese and English scores and truncate the result to an integer.

    【Answer】=INT(AVERAGE(B2:C2)) ➔ 86

DSE Bullet Series – Database and SQL (Solutions 14)

1. STUDENT Table #Read the data before reading the SQL

NULL means that no score was entered for that student. A query result shows only the columns named in SELECT.

STUID	SNAME	CLASS	TEST1	TEST2
S01	Siu-mei	4A	82	91
S02	Siu-yat	4A	65	58
S03	Ka-fai	4B	74	88
S04	Ka-ching	4B	90	76
S05	Hiu-yan	4C	58	83
S06	Siu-tung	4C	77	94
S07	Hardy	4A	95	NULL
S08	Chi-ching	4B	45	62

2. Reading SQL #Start with FROM and narrow the records step by step

FROM: choose the table. WHERE: keep only records that satisfy the condition; put text in quotation marks and test NULL with IS NULL.

SELECT: output only the named columns; * means all columns and DISTINCT removes duplicates.

GROUP BY: group records by a column; COUNT, SUM, AVG, MAX and MIN then return one result for each group.

HAVING: filter groups. ORDER BY: sort the final result; ASC means ascending and DESC means descending.

*The Core DSE paper mainly asks you to read SQL and draw the output. This pack practises that first and leaves writing SQL to the final page.

3. Worked Examples: SQL → Output #Study two complete examples first

Example 1:

```
SELECT SNAME, TEST1 FROM STUDENT WHERE CLASS = '4A' ORDER BY TEST1
DESC;
```

SNAME	TEST1
Hardy	95
Siu-mei	82
Siu-yat	65

Example 2:

```
SELECT CLASS, COUNT(*) AS TOTAL FROM STUDENT GROUP BY CLASS;
```

CLASS	TOTAL
4A	3
4B	3
4C	2

4. Read SQL and Draw the Output (1/2) #Core DSE style: read the SQL only

Using the STUDENT table, write the output of each SQL query in the box.

1.

```
SELECT SNAME FROM STUDENT WHERE CLASS = '4A';
```

SNAME
Siu-mei
Siu-yat
Hardy

2.

```
SELECT SNAME, TEST1 FROM STUDENT WHERE TEST1 >= 70 ORDER BY TEST1
DESC;
```

SNAME	TEST1
Hardy	95
Ka-ching	90
Siu-mei	82
Siu-tung	77
Ka-fai	74

3.

```
SELECT SNAME FROM STUDENT WHERE TEST1 > 90 OR TEST2 > 90;
```

SNAME
Siu-mei
Hardy

4.

```
SELECT SNAME FROM STUDENT WHERE TEST1 IS NULL;
```

SNAME
Hardy

5.

```
SELECT DISTINCT CLASS FROM STUDENT ORDER BY CLASS;
```

CLASS
4A
4B
4C

4. Read SQL and Draw the Output (2/2) #Check GROUP BY and sorting carefully

6.

```
SELECT CLASS, COUNT(*) AS TOTAL FROM STUDENT GROUP BY CLASS;
```

CLASS	TOTAL
4A	3
4B	3
4C	2

7.

```
SELECT CLASS, MAX(TEST2) AS HIGHEST FROM STUDENT GROUP BY CLASS ORDER
BY HIGHEST DESC;
```

CLASS	HIGHEST
4C	94
4A	91
4B	88

8.

```
SELECT SNAME FROM STUDENT WHERE TEST1 BETWEEN 60 AND 80 ORDER BY
TEST1;
```

SNAME
Siu-yat
Ka-fai
Siu-tung

9.

```
SELECT SNAME FROM STUDENT WHERE SNAME LIKE 'Siu-%' ORDER BY TEST1
DESC;
```

SNAME
Siu-mei
Siu-tung
Siu-yat

10.

```
SELECT CLASS, SUM(TEST2) AS TOTAL2 FROM STUDENT GROUP BY CLASS;
```

CLASS	TOTAL2
4A	149
4B	226
4C	177

5. Write SQL Yourself #Extension: work backwards from the output

These questions go beyond the Core paper, but they help you become familiar with SQL structure.

1. List the names and TEST1 scores of 4A students, with the highest TEST1 score first.

   【Answer】SELECT SNAME, TEST1 FROM STUDENT WHERE CLASS = '4A' ORDER BY TEST1 DESC;

2. List the names of students whose TEST1 score is below 60.

   【Answer】SELECT SNAME FROM STUDENT WHERE TEST1 < 60;

3. Count the number of students in each class.

   【Answer】SELECT CLASS, COUNT(*) AS TOTAL FROM STUDENT GROUP BY CLASS;

4. Find the highest TEST2 score in each class.

   【Answer】SELECT CLASS, MAX(TEST2) AS HIGHEST FROM STUDENT GROUP BY CLASS;

5. List the names of students who do not have a TEST1 score.

   【Answer】SELECT SNAME FROM STUDENT WHERE TEST1 IS NULL;

DSE Bullet Series – Spreadsheet Data Analysis (Solutions 16)

1. Past Questions (2015–2021) #Identify the purpose, then choose the tool

Year	Past Question	Focus / Marks
2015	Commission and what-if analysis	Input cells can be changed; output changes with input
2016	Daily net profit; pivot table	Rows, columns, values; chart type and chart features
2017	Convert a long table into a pivot table	Row label, column label, value field and summary function
2021	Sort by subject ascending and score descending	Primary/secondary sort keys; rank

2. Key Notes #Sorting, filtering and pivot tables have different purposes

Sorting: several sort keys can be specified; records are sorted by the primary key first and by the secondary key only when the primary values are equal. Ascending means smallest to largest / A to Z; descending is the reverse. Sorting changes the order of records.

Filtering: only records that meet the condition are displayed while the other rows are temporarily hidden, not deleted; clearing the filter displays them again. Use AND when all conditions must hold and OR when any one may hold.

Pivot table: groups and summarises a large number of records. Row labels appear on the left, column labels across the top, and a value field can use sum, average or count. Refresh the pivot table after the source data change.

Charts: use a column chart or bar chart to compare items, a line chart to show a trend and a pie chart to show the proportion of each part. Select the data range and chart type, then add a title, axis labels, legend and data labels as appropriate.

What-if analysis: change an input cell and observe the result in an output cell; first identify which inputs the formula depends on, then decide which cell may be changed.

*Common mistakes: treating filtering as deletion; reversing the order of a multi-level sort; placing values in row or column labels in a pivot table; using a pie chart to compare items; changing a formula instead of an input in what-if analysis.

3. Data Table #The questions below refer to this table

A1:D9; row 1 contains headings and rows 2–9 contain the scores of four students in two subjects.

Class	Name	Subject	Score
Class A	Mei	Chinese	82
Class A	Mei	English	91
Class A	Ka-fai	Chinese	74
Class A	Ka-fai	English	88
Class B	Siu-tung	Chinese	77
Class B	Siu-tung	English	94
Class B	Ka-ching	Chinese	90
Class B	Ka-ching	English	76

4. Similar Questions #A different setting, the same pattern

1. Refer to the data table in Section 3 and complete the following.

(a) Sort the records by Subject ascending and Score descending. Write the names and scores in the first **three** rows after sorting.

【Answer】 Chinese: Ka-ching 90, Mei 82, Siu-tung 77; English: Siu-tung 94, Mei 91, Ka-fai 88. Therefore the first three rows are Ka-ching 90, Mei 82 and Siu-tung 77.

(b) Filter for Subject = English and Score >= 85. State the names displayed and explain the difference between filtering and deleting.

【Answer】 Siu-tung (94) and Mei (91) are displayed. Filtering only temporarily hides records that do not meet the condition; it does not delete them, and clearing the filter displays them again.

2. The school creates a pivot table from the data table, with Class as row labels, Subject as column labels and the sum of Score as the value.

(a) Complete the pivot table below.

Class	Chinese	English	Total
Class A	156	179	335
Class B	167	170	337
Total	323	349	672

(b) If Class is moved to column labels and Subject to row labels, will the totals change? Explain.

【Answer】 No. Changing the layout only rearranges how the same values are displayed; it does not change the source data or the summarised values, so the totals remain the same.

(c) To compare the total English scores of Class A and Class B, suggest a chart type and give one reason.

【Answer】 A column chart; it places the two categories (Class A and Class B) side by side so that the difference in their total English scores is easy to compare.